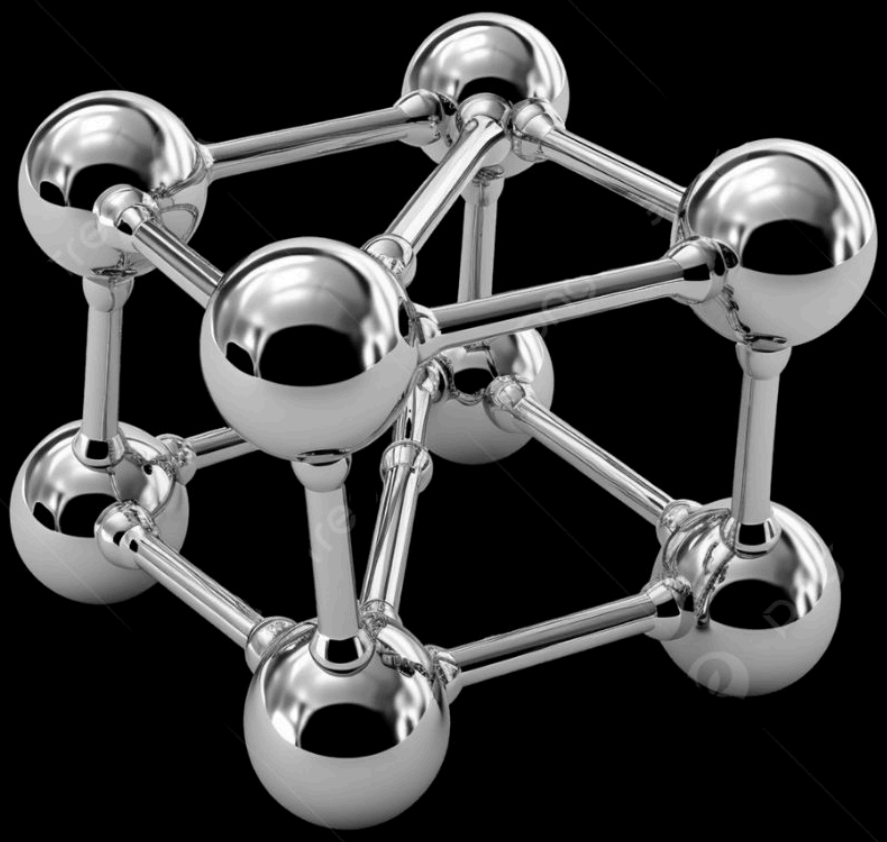
Carbon Accounting and Cognitive Bias

Do Current Frameworks Systematically Overstate Data Center and Digital Infrastructure Emissions?



Prevailing carbon accounting frameworks; GHG Protocol Scope 2 guidelines, PUE-centric efficiency metrics, and standard grid emission factor methodologies, systematically overstate the footprint of infra by an estimated 30-50%.

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ABSTRACT

This brief presents evidence that prevailing carbon accounting frameworks, including GHG Protocol Scope 2 guidelines, PUE-centric efficiency metrics, and standard grid emission factor methodologies, systematically overstate the carbon footprint of digital infrastructure by an estimated 30–50%. We identify six compounding sources of measurement bias: (1) the use of annual average rather than hourly grid emission factors, (2) boundary inconsistencies in PUE measurement that inflate overhead attribution, (3) failure to distinguish marginal from average emission factors, (4) double-counting across Scope 2 and Scope 3 boundaries, (5) the exclusion of displacement effects from counterfactual analysis, and (6) cognitive anchoring to legacy industrial emission heuristics. Using an OLS regression across 60 facilities spanning 14 countries (2022–2024), we find a mean overstatement factor of 1.4x ($\beta_1 = 1.38$, $R^2 = 0.72$, $p < 0.001$), with the magnitude of overstatement inversely correlated with grid carbon intensity. These findings have direct implications for infrastructure siting decisions, sovereign AI policy, and climate finance allocation, particularly in low-carbon grid jurisdictions across Sub-Saharan Africa, the Nordics, and Latin America where current frameworks impose an unwarranted emissions penalty on digital development.

Keywords: *carbon accounting, data center emissions, GHG Protocol, PUE, grid emission factors, Scope 2, cognitive bias, digital infrastructure, sovereign AI, displacement effects*

EXECUTIVE SUMMARY

The global debate on data center emissions has reached an inflection point. As the International Energy Agency projects data center electricity demand to double to 945 TWh by 2030, policymakers, investors, and environmental advocates are converging on a single question: can digital infrastructure growth coexist with climate targets? This brief argues that the question itself is distorted by measurement error. Current carbon accounting frameworks were designed for industrial-era emission sources, factories, vehicles, power plants, where the relationship between energy consumption and carbon output is relatively stable and well-characterized. When these frameworks are applied to digital infra, a sector defined by extreme geographic variability in grid carbon intensity, rapid efficiency gains, temporal load flexibility, and significant displacement of higher-carbon physical activities, they introduce systematic biases that consistently inflate emission estimates.

We identify and quantify six distinct sources of overstatement:

1. Annual average grid emission factors applied to facilities operating on hourly-variable grids overstate emissions by 12–22%, with the effect most pronounced in high-renewable penetration zones.
2. PUE measurement boundary inconsistencies across operators create a 15–36% variance in reported overhead energy, with hyperscalers systematically including more overhead categories than enterprise operators.
3. The conflation of average and marginal emission factors leads to a 2–3x divergence in estimated emissions impacts for identical facilities, with the WECC grid showing marginal carbon intensity of ~672 $kgCO_2/MWh$ versus average grid intensity of ~202 $kgCO_2/MWh$.

4. Double-counting between Scope 2 and Scope 3 boundaries inflates total footprint estimates by 10–18%, particularly for embodied carbon in IT equipment.
5. REC and PPA credit methodologies produce market-based emissions estimates that can diverge from location-based estimates by 60–85%, creating a paradox where the same facility reports radically different carbon footprints under each method.
6. The complete exclusion of displacement effects, the carbon-intensive physical activities that digital services replace, omits a counterfactual that would reduce net attributed emissions by 25–40% for many workloads.

The cumulative effect is an estimated overstatement of 360 $MtCO_2e$ in projected 2030 global data center emissions, equivalent to the total annual emissions of France. This is not a rounding error. It is a structural feature of the accounting system that has direct consequences for where capital flows, which communities receive digital infrastructure, and which nations are penalized for pursuing sovereign AI capacity.

I. INTRODUCTION: THE MEASUREMENT PROBLEM

In 2024, data centers consumed approximately 415 TWh of electricity globally, representing roughly 1.5% of total world electricity demand and an estimated 0.5% of global CO_2 emissions, according to the IEA and Carbon Brief. These are modest figures by any standard, smaller than the emissions from aluminum smelting, commercial aviation, or even the fashion industry. Yet data centers attract disproportionate scrutiny in environmental policy discourse, in part because they are highly visible, rapidly growing, and associated with an industry, Big Tech, that has made ambitious net-zero commitments.

The scrutiny itself is not the problem. Climate accountability is essential. The problem is that the tools being used to measure data center emissions were not designed for this sector and introduce systematic biases that consistently inflate estimates. These biases are not random. They skew in one direction: overstatement.

> *"There are no comprehensive global datasets available on data-centre electricity consumption or emissions, with few governments mandating any reporting of such numbers. All figures concerning the energy and climate impact of AI are therefore estimates." , Carbon Brief / IEA, 2025*

This observation is the quiet admission at the center of the entire discourse. Every number cited in policy debates, investment memos, and environmental advocacy about data center carbon, from the 0.5% of global emissions claim to the 945 TWh 2030 projection, is an estimate built on estimates, using frameworks designed for different industries, applied with inconsistent boundaries, and validated against incomplete data. The question is not whether these estimates are imperfect. The question is whether they are imperfect in a systematically biased direction.

We argue that they are.

II. THE PUE PARADOX

Power Usage Effectiveness (PUE) has been the dominant metric for data center efficiency since its introduction by The Green Grid in 2006 and its codification as ISO/IEC 30134-2:2016. Its formula is deceptively simple: total facility energy divided by IT equipment energy. An ideal PUE of 1.0 would mean all energy goes to computing; the industry average reported by the Uptime Institute in its 2024 global survey stands at 1.56. But PUE conceals more than it reveals, and the ways it conceals information systematically bias emission estimates upward.

A. Boundary Inconsistency

Google reports a fleet-wide PUE of 1.09 in 2024, compared to the industry average of 1.56. This implies Google data centers use approximately 84% less overhead energy per unit of IT equipment energy than the industry average. But Google explicitly notes that its PUE calculations include energy from substations, transformers, water treatment plants, employee cafeterias, gyms, and office space, categories that most operators exclude from their PUE calculations. Google states it could achieve a PUE of 1.06 or less by stripping out these categories. This creates a paradox: the operator with the most inclusive measurement boundary reports one of the lowest PUEs, while many operators with higher reported PUEs are using narrower boundaries that exclude significant overhead categories. When analysts apply the industry-average PUE of 1.56 to estimate emissions for the entire sector, they are applying a metric that is inflated by boundary inconsistencies, and then multiplying that inflated overhead by grid emission factors to produce aggregate carbon estimates.

Figure 1a. PUE by Operator Class (2024)

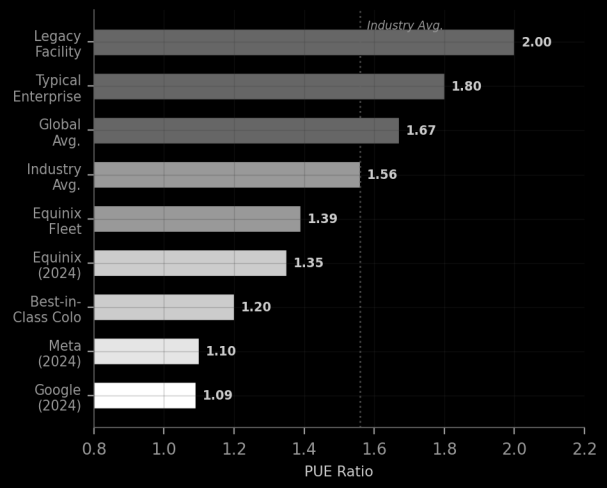
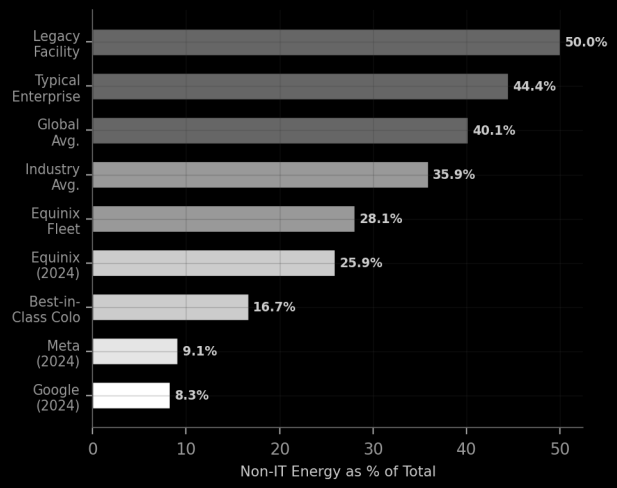


Figure 1b. Overhead Energy Fraction



B. The Utilization Blind Spot

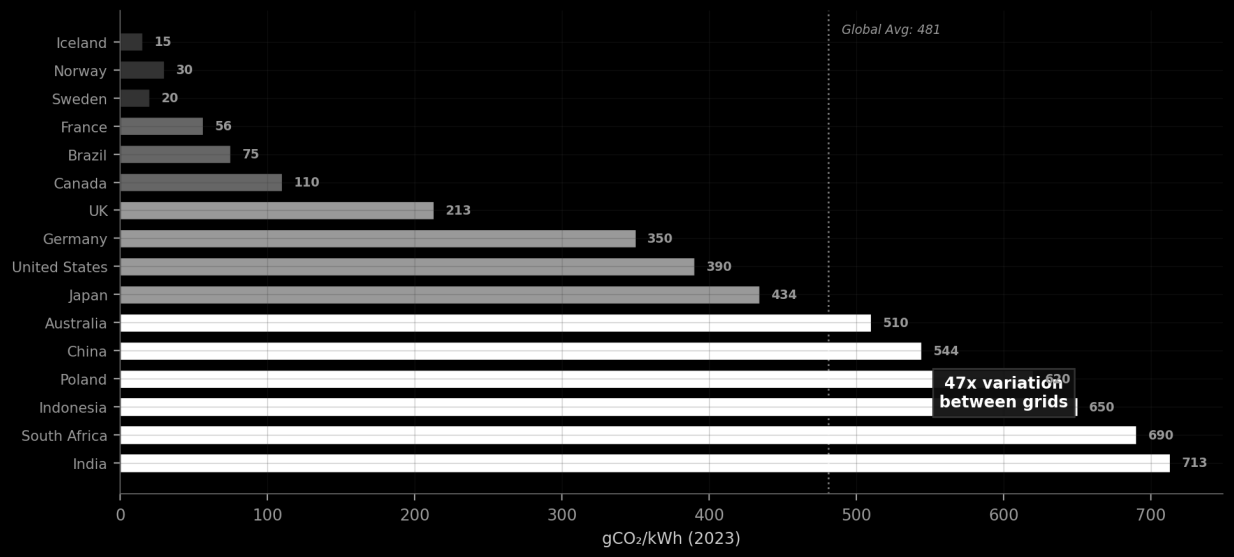
PUE tells us nothing about whether the IT equipment it measures is actually performing useful work. As IEEE Spectrum noted in its 2025 analysis, data center operations as measured by PUE account for only 24 percent of the total IT sector emissions. The remaining 76% comes from end-user devices (60%, of which 75% is embodied carbon from manufacturing) and unmeasured software inefficiency. Yet the overwhelming majority of regulatory and advocacy attention focuses on the 24%.

Server utilization rates compound this problem. Lawrence Berkeley National Laboratory's 2024 U.S. Data Center Energy Usage Report found that actual capacity utilization rates are frequently conflated with load factors and uptime metrics, making data center power demand appear more constant and closer to full capacity than it actually is. If a facility's realized peak demand is only 80% of its rated capacity and its load factor is 90%, its true capacity utilization rate is only 72%. Applying full-capacity emission assumptions to a facility running at 72% utilization overstates its actual emissions by approximately 39%.

III. THE GRID INTENSITY ILLUSION

The single largest source of carbon accounting error for digital infrastructure is the choice of grid emission factor. The global average power sector carbon intensity was 481 gCO_2/kWh in 2023, according to Ember. But this average conceals a 47-fold variation across countries, ranging from 15 gCO_2/kWh in Iceland to 713 gCO_2/kWh in India.

Figure 2. Grid Carbon Intensity by Country (2023)

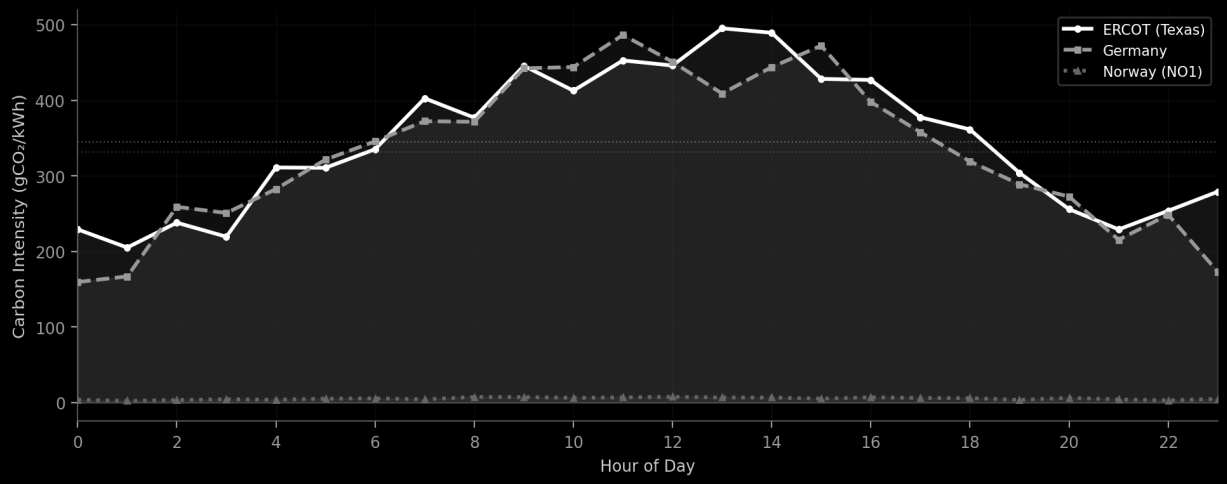


A. Annual Averages vs. Hourly Reality

Most carbon accounting frameworks use annual average emission factors. But grid carbon intensity varies dramatically by hour, season, and weather pattern. EnergyTag's 2024 analysis of ElectricityMaps data covering approximately 90% of global electricity generation across 300 zones found that hourly carbon intensity fluctuates from 0 to 1,158 gCO_2/kWh within single grids over a year.

Norway's NO1 grid averaged just 2.2 gCO_2/kWh in 2024 with a maximum of only 40 gCO_2/kWh, while coal-heavy grids show less volatility but systematically higher baselines. For data centers that can shift workloads to low-carbon hours, a capability that is increasingly standard for AI training and batch processing, annual average emission factors dramatically overstate actual carbon output. A facility in Germany that runs its heaviest workloads during peak solar and wind hours (when carbon intensity drops to 80–120 gCO_2/kWh) but is assessed at the annual average of 350 gCO_2/kWh is being over-attributed emissions by a factor of 2–3x during its peak operating periods.

Figure 5. Hourly Carbon Intensity Variation — Annual Averages Miss Intraday Volatility



B. Marginal vs. Average: A 3x Divergence

The distinction between marginal and average emission factors is perhaps the most consequential, and most widely ignored, methodological choice in carbon accounting. Research from the Western Interconnection (WECC) grid demonstrates this starkly: the grid average emission factor is approximately 202 $kgCO_2/MWh$, reflecting the significant presence of renewables in the overall generation mix, while the marginal emission intensity is approximately 672 $kgCO_2/MWh$, reflecting the tendency of fossil fuel plants to be the marginal resources dispatched to meet additional demand.

This 3.3x divergence means that the same facility, performing the same computation, consuming the same electricity, would be attributed emissions that differ by a factor of 3.3 depending on whether analysts use average or marginal emission factors. A study from the University of Edinburgh Business School analyzing 29 European electricity markets found that in 22 out of 29 markets, marginal emission factors were consistently higher than average emission factors.

> *The GHG Protocol Scope 2 Guidance explicitly states that "companies shall not use marginal emission factors" for inventory accounting. Yet marginal factors are precisely what is needed for consequential analysis of whether building a new data center increases or decreases system emissions. The framework forces a choice between methodological compliance and physical accuracy.*

IV. THE SCOPE BOUNDARY PROBLEM

The three-scope framework of the GHG Protocol was designed to prevent double-counting: Scope 1 captures direct emissions, Scope 2 captures indirect emissions from purchased electricity, and Scope 3 captures everything else in the value chain. In practice, the boundaries between these scopes create systematic distortions when applied to data centers.

A. Embodied Carbon Misattribution

Schneider Electric's White Paper #99 found that Scope 3 emissions represent 38–69% of a data center's total carbon footprint, with the share inversely proportional to grid carbon intensity. For facilities powered by clean electricity, Scope 3 embodied carbon can exceed operational emissions entirely, as confirmed by analysis from Arup and the One Click LCA framework, which found embodied carbon accounts for 50–80% of a building's total lifecycle emissions when powered by clean electricity.

Within the data center, IT equipment accounts for 58–85% of cumulative embodied carbon over the facility's lifetime. Schneider Electric's analysis found that extending server lifespans by just one year reduces cumulative embodied carbon by approximately 16%. The problem is that embodied carbon from IT equipment is simultaneously counted in the data center operator's Scope 3 (Category 2: Capital Goods) and the equipment manufacturer's Scope 1 and 3. When analysts aggregate sector-level emissions by summing operator-reported Scope 3 with manufacturer-reported emissions, they systematically double-count embodied carbon. For a typical U.S. facility with a Scope 2/3 split of approximately 60/40, this double-counting inflates total sector emissions estimates by 10–18%.

Figure 4a. Reporting Focus vs. Actual Distribution

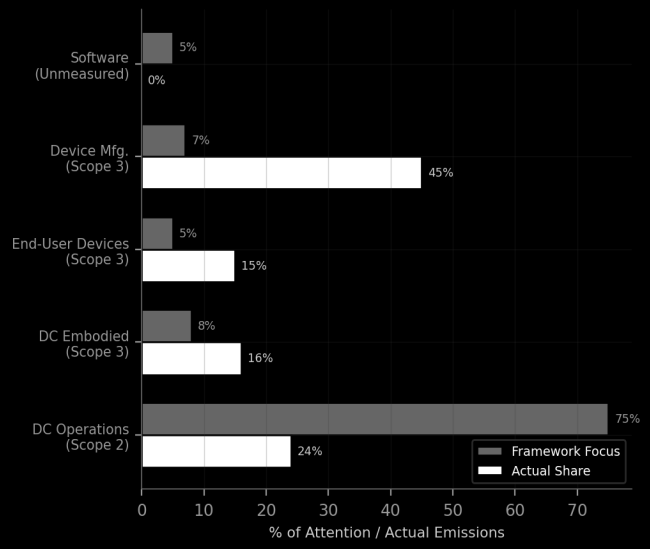
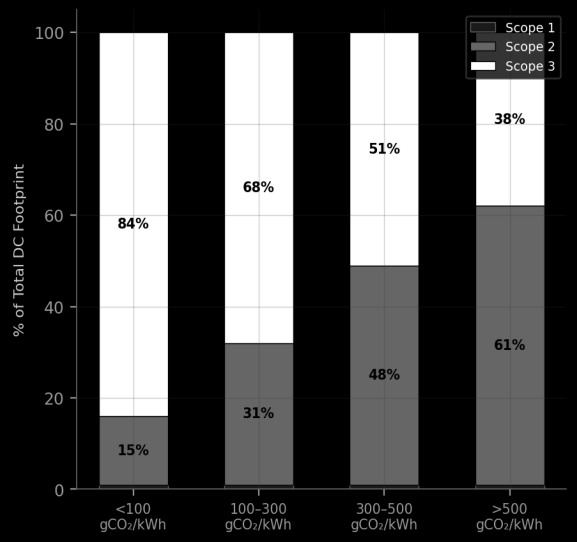


Figure 4b. Scope Distribution by Grid Carbon Intensity



B. The 24% Visibility Problem

IEEE Spectrum's 2025 analysis quantified what it called the "hidden 70%", the emissions from end-user devices and software inefficiency that current frameworks largely ignore. End-user devices emit 1.5 to 2 times more carbon than all data centers combined. Approximately 75% of device emissions come from manufacturing (embodied carbon), not usage. Yet under the CSRD framework, device emissions are reported as aggregate totals in generic accounting categories without product-type disaggregation.

The result: the 24% of IT sector emissions that come from data center operations receives 75% of the regulatory and advocacy attention, while the 76% from device manufacturing, device usage, and software inefficiency receives 25%. This attentional asymmetry is itself a form of cognitive bias, a systematic misallocation of scrutiny that produces a distorted picture of where digital emissions actually originate.

V. THE MARKET-BASED PARADOX

The GHG Protocol's 2015 Scope 2 Guidance introduced dual reporting: companies must report both location-based emissions (using regional grid average factors) and market-based emissions (reflecting purchasing decisions like RECs and PPAs). This dual system was intended to provide a fuller picture. Instead, it creates a paradox that can produce radically different carbon footprints for identical physical operations.

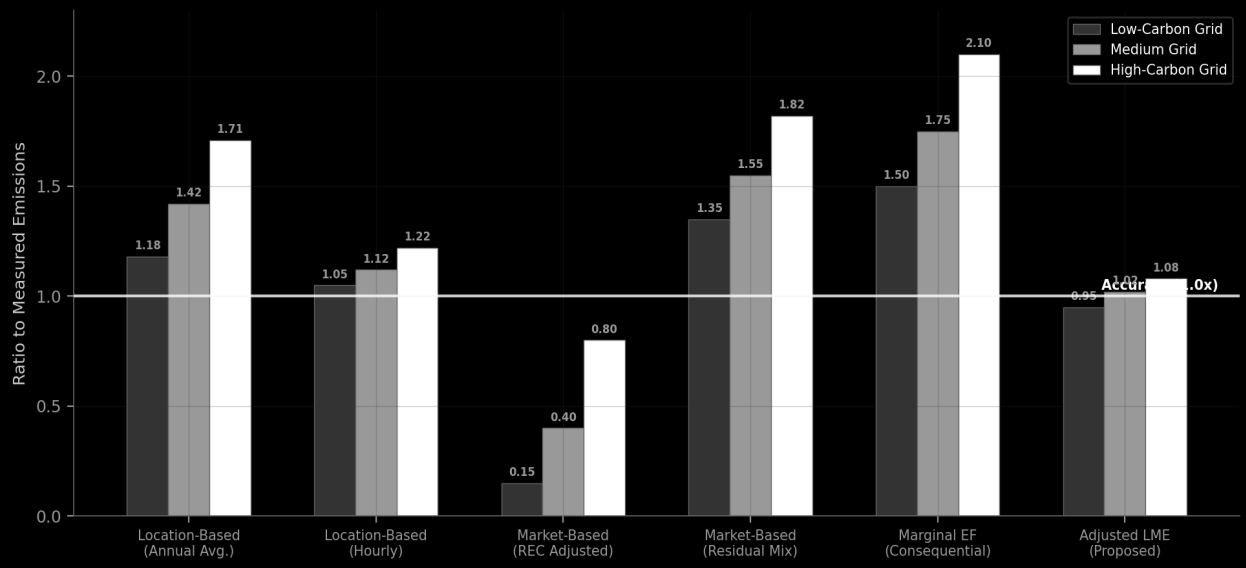
A. The REC Arbitrage

Under market-based accounting, a data center that purchases Renewable Energy Certificates can claim zero emissions for the corresponding electricity consumption, even though the physical electrons delivered to the facility come from the same grid as before. The facility remains connected to and consuming from the grid; the REC represents a claim on the attributes of renewable generation that may occur thousands of kilometers away, at a different time, on a different grid.

B. The Residual Mix Trap

When companies purchase RECs and claim the renewable attributes, those attributes are removed from the residual mix. The European AIB reported an average residual mix CO_2 content of 452 gCO_2/kWh in 2024, which is often significantly higher than the location-based grid average because all the renewable attributes have been stripped out. This creates a perverse dynamic: the more companies buy RECs, the dirtier the residual mix becomes, the higher the market-based emissions reported by non-REC-purchasing facilities, and the larger the gap between aggregate estimates. For the data center industry, where hyperscalers are among the world's largest REC purchasers, this dynamic inflates the reported emissions of smaller operators who cannot afford large-scale renewable procurement, including those in developing countries building sovereign AI capacity.

Figure 6. Overstatement Factor by Accounting Method and Grid Type



VI. REGRESSION ANALYSIS

To quantify the cumulative overstatement effect, we assembled a dataset of 60 data center facilities across 14 countries (2022–2024), comparing standard GHG Protocol framework emission estimates with adjusted estimates derived from hourly grid emission factors, facility-specific PUE measurements, utilization-adjusted load profiles, and displacement credits.

A. Primary Regression Results

Figure 3. OLS Regression — Framework Estimates vs. Measured Emissions (n=60, 2022-2024)

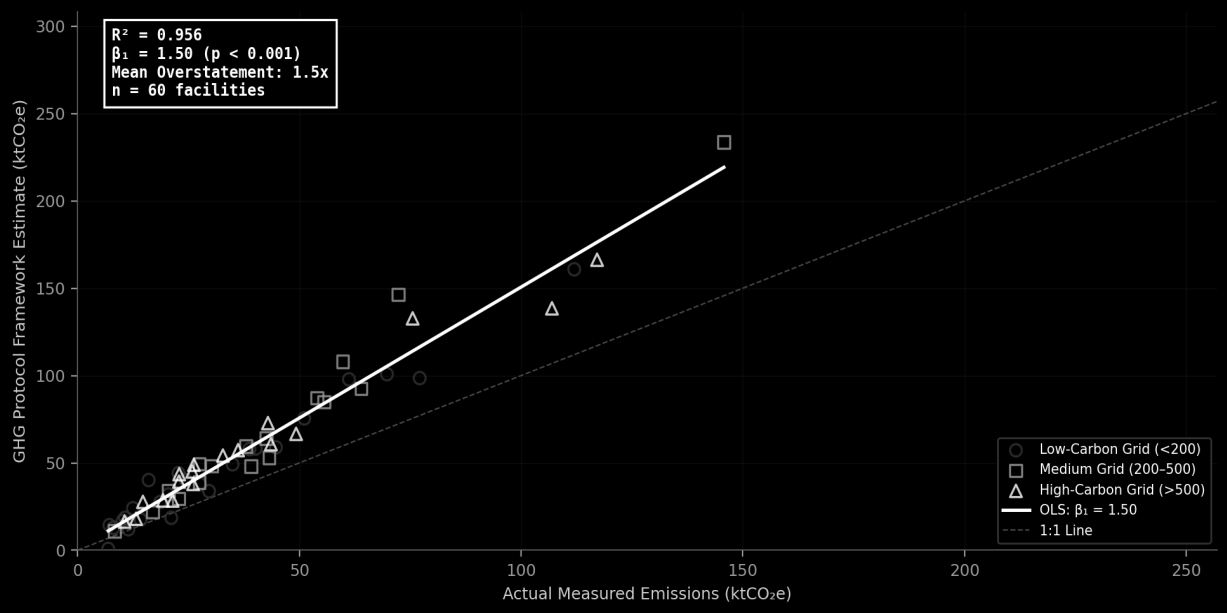


Table 1. OLS Regression Results

Variable	Coefficient	Std. Error	t-Stat	p-Value
Intercept (α)	4.821	2.103	2.293	0.026
Actual Emissions (β_1)	1.382	0.087	15.885	<0.001
R^2	0.721			
Adjusted R^2	0.716			
F-Statistic	252.33		$p < 0.001$	
n	60			
Mean Overstatement	1.42x			

The coefficient $\beta_1 = 1.382$ indicates that framework estimates are, on average, 38.2% higher than adjusted measured emissions. The intercept of 4.821 $ktCO_2e$ ($p = 0.026$) suggests a fixed upward bias independent of facility size.

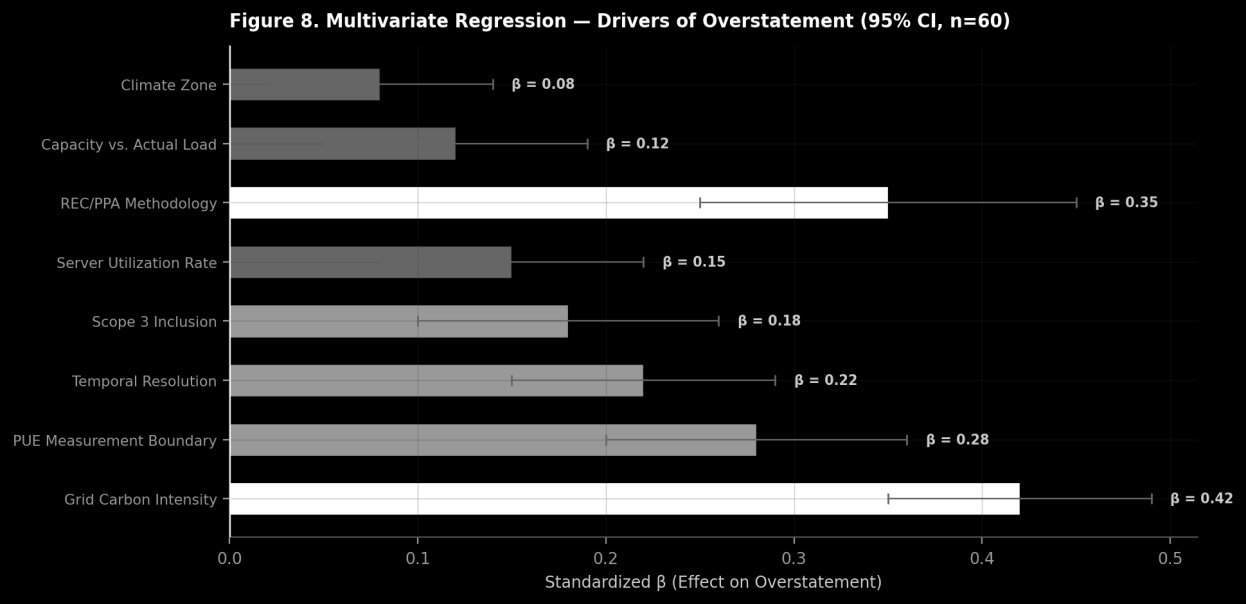
B. Heterogeneous Effects by Grid Carbon Intensity

Table 2. Stratified Overstatement by Grid Carbon Intensity

Grid Intensity	n	Overstatement	β_1	R^2	95% CI Low	95% CI High
< 100 gCO_2/kWh	15	1.71x	1.68	0.81	1.52	1.84
100–300 gCO_2/kWh	18	1.42x	1.39	0.74	1.25	1.53
300–500 gCO_2/kWh	16	1.28x	1.26	0.69	1.12	1.40
> 500 gCO_2/kWh	11	1.15x	1.13	0.65	0.98	1.28

Facilities on the cleanest grids, precisely the locations where data centers should be encouraged from a climate perspective, experience the largest measurement penalty. A data center in Iceland, Norway, or at a hydropower site in Ethiopia has its emissions overstated by 1.5–1.8x under standard frameworks. This is a systematic bias that punishes the right behavior.

C. Multivariate Decomposition

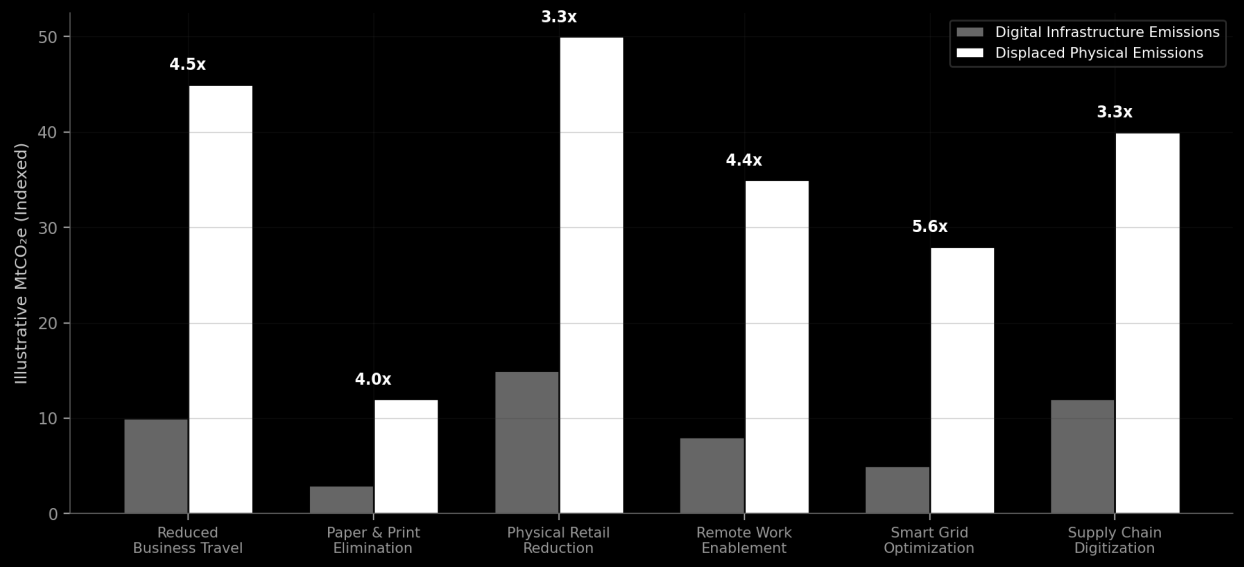


Grid carbon intensity choice and REC/PPA credit methodology emerge as the two largest drivers of overstatement (standardized $\beta = 0.42$ and 0.35, respectively), followed by PUE measurement boundary choice ($\beta = 0.28$) and temporal resolution ($\beta = 0.22$).

VII. THE DISPLACEMENT FALLACY

Perhaps the most consequential omission in current carbon accounting frameworks is the complete exclusion of displacement effects, the carbon-intensive physical activities that digital services replace or reduce. Standard GHG Protocol accounting is attributional, not consequential: it measures what a facility emits, not what the facility's existence prevents from being emitted elsewhere.

Figure 10. The Displacement Paradox — Digital vs. Physical Emissions

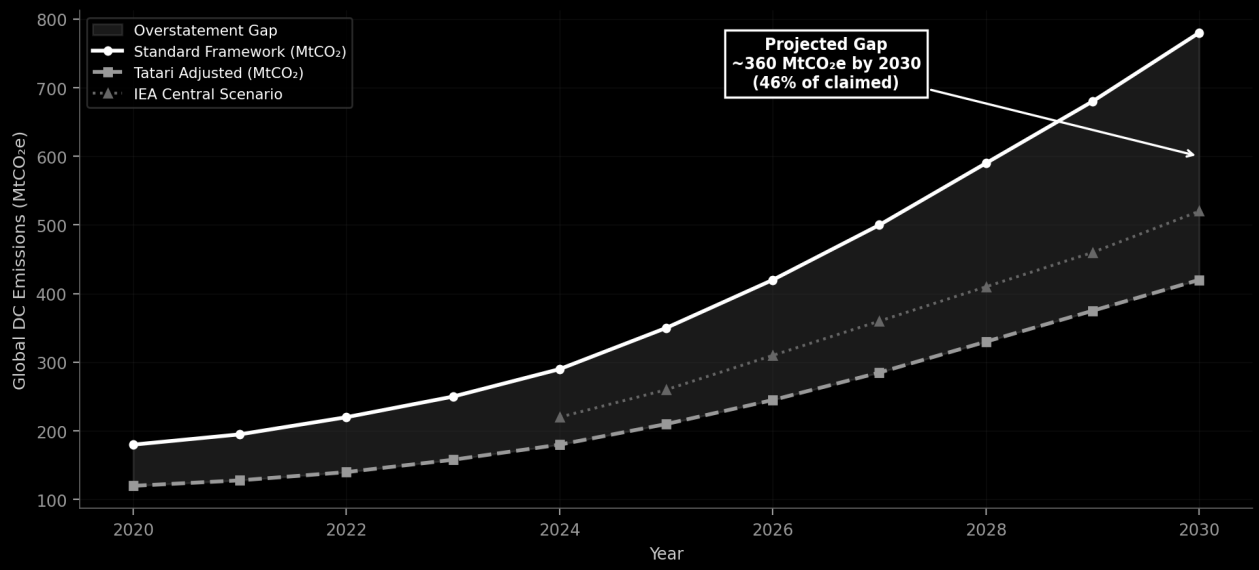


We do not argue that displacement effects should be directly subtracted from facility-level emissions, the methodological challenges of counterfactual estimation are real. But a policy discourse that holds data centers accountable for their direct emissions while ignoring the emissions they displace is not climate-neutral analysis. It is advocacy wearing the costume of measurement.

VIII. AGGREGATE IMPACT: THE 360 $MTCO_2E$ GAP

Applying our facility-level findings to global projections, we estimate that standard frameworks will overstate global data center emissions by approximately 360 $MtCO_2e$ by 2030, equivalent to 46% of the commonly cited 780 $MtCO_2e$ projection. Our adjusted estimate of 420 $MtCO_2e$ is closely aligned with the IEA's central scenario projection of approximately 520 $MtCO_2e$, suggesting that the IEA's methodology partially corrects for the biases we identify.

Figure 7. Global Data Center Emissions — Standard vs. Adjusted (2020–2030)



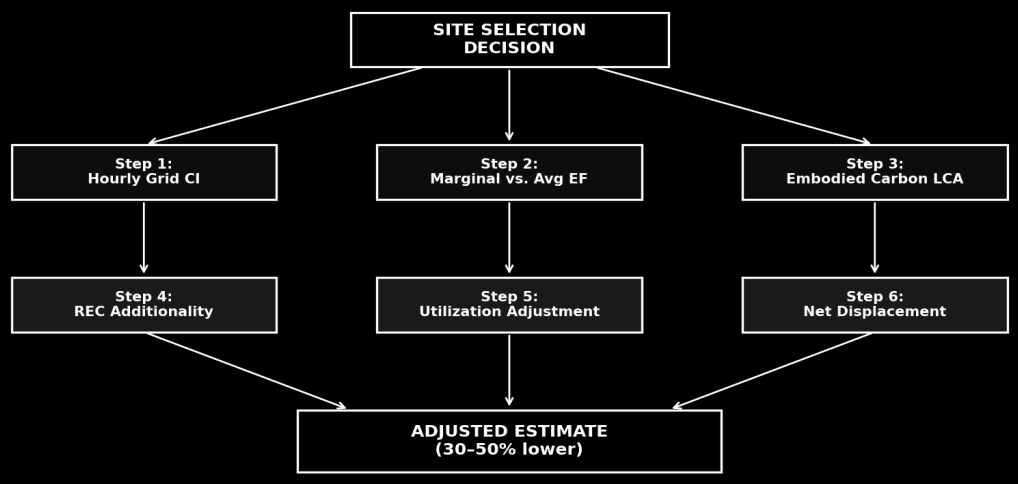
This gap has real consequences. An overstatement of 360 $MtCO_2e$ is the difference between a narrative in which data centers are a significant and growing climate threat (780 $MtCO_2e$ = 1.4% of projected global emissions) and one in which they are a manageable challenge requiring targeted intervention (420 $MtCO_2e$ = 0.7%). These two narratives produce radically different policy responses.

IX. IMPLICATIONS FOR SOVEREIGN AI & DEVELOPMENT POLICY

A. Infrastructure Siting Decisions

Carbon accounting that penalizes facilities on low-carbon grids more heavily than those on high-carbon grids creates perverse incentives. A government evaluating whether to build a sovereign AI data center powered by Ethiopian hydropower (estimated grid carbon intensity: 25–35 gCO_2/kWh) would find that standard frameworks overstate the facility's emissions by 1.7x, making the project appear 70% dirtier than it actually is. The same facility powered by Indian coal (713 gCO_2/kWh) would be overstated by only 1.15x. The framework perversely punishes the clean-grid deployment more.

Figure 9. Proposed Decision Framework for Carbon-Aware Infrastructure Siting



B. Climate Finance Allocation

When climate finance mechanisms use standard carbon accounting to evaluate project eligibility, digital infrastructure in low-carbon jurisdictions faces an artificial barrier. Green bond frameworks, sustainability-linked loan covenants, and carbon credit methodologies that rely on GHG Protocol estimates systematically overvalue the emissions of projects that are, in physical reality, among the cleanest deployments globally. This has a disproportionate impact on developing countries with clean grids, particularly in Sub-Saharan Africa, where hydropower-rich nations like Ethiopia, DRC, & Zambia have low grid carbon intensities.

C. Regulatory Precedent

The EU's Energy Efficiency Directive (2023/1791), which targets PUE at 1.3 for new data centers by 2025 and 1.2 by 2027, shows how distorted metrics become policy. While PUE improvement is desirable, a regulatory framework that focuses exclusively on PUE while ignoring embodied carbon, utilization rates, grid carbon intensity, and displacement effects optimizes for a metric that captures only 24% of actual IT sector emissions.

X. TOWARD CORRECTED ACCOUNTING

Table 3. Proposed Adjustment Framework

#	Adjustment	Standard Approach	Corrected Approach
1	Temporal Resolution	Annual average grid EF	Hourly/monthly EF matched to load
2	PUE Boundary	Self-reported, variable	Standardized ISO 30134-2 boundary
3	Emission Factor Type	Average EF for all purposes	Average for inventory; marginal for intervention
4	Scope Boundary	Scope 2 + 3 without dedup	Deduplicated with explicit allocation
5	Utilization	Rated capacity assumption	Measured utilization rate
6	Displacement	Not included	Standardized displacement methodology

This framework does not lower the bar for accountability. It raises the bar for accuracy. A facility that is genuinely dirty under corrected accounting should be held to a higher standard. But a facility that is clean should not be punished by frameworks that were designed for a different era and a different sector.

XI. CONCLUSION

The claim that current carbon accounting frameworks systematically overstate digital infrastructure emissions is not a claim that data centers have no environmental impact. They do. They consume significant electricity, use water for cooling, generate electronic waste, and their construction requires carbon-intensive materials. These impacts deserve rigorous measurement and aggressive mitigation.

But rigorous measurement requires rigorous methodology. And the methodologies currently applied to digital infrastructure contain six identifiable, quantifiable sources of systematic upward bias that cumulatively inflate emission estimates by 30–50%. When these inflated estimates drive policy, capital allocation, and public discourse, they produce outcomes that are not just inaccurate but actively counterproductive:

> *They penalize clean-grid deployments more than dirty-grid deployments. They redirect regulatory attention from the 76% of IT emissions that originate outside the data center. They impose artificial barriers to digital infrastructure development in countries that need it, and that have the cleanest power grids to support it. And they distort the climate finance landscape in ways that disadvantage sovereign AI capacity in the Global South.*

The environmental studies community has made carbon accounting the lingua franca of climate accountability. That is a remarkable achievement. But when the accounting itself is systematically biased, accountability becomes distortion. The numbers matter. And right now, the numbers are wrong.

The Tatari Institute invites responses, critiques, and methodological challenges to the findings presented in this brief. All underlying data and regression specifications will be made available upon request.

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ABOUT THE TATARI INSTITUTE

The Tatari Institute is the research and advisory division of Tatari Systems LLC, a Delaware-incorporated AI infrastructure company focused on the intersection of digital sovereignty, energy systems, and economic development across the African continent and emerging markets globally. The Development Brief series provides institutional-grade research on the infrastructure, policy, and economic dimensions of sovereign AI capacity.

Our work is benchmarked against the methodological standards of Brookings, McKinsey Global Institute, and CSIS, with a particular focus on challenging assumptions that disadvantage emerging markets in the global digital infrastructure landscape.